If quantum field theory describes the world as a wave function of a classical field configuration, that seems to be waviness on top of waviness. If we asked how much wavier things could possibly get, the answer (to paraphrase Nigel Tufnel of Spinal Tap) might be “none more wavy.” And yet, when we make observations of quantum fields, for example, in a detector at the Large Hadron Collider in Geneva, what we see are individual tracks representing the paths of point-like objects, not diffuse wavy clouds. Somehow we have circled back to particles, despite being as wavy as can be.

The reason for this goes back to the same reason why we see discrete energy levels for electrons in atoms. An electron moving through space all by itself can have any energy at all, but in the vicinity of the attractive force exerted by an atomic nucleus, it’s as if the electron is trapped in a box. The wave function falls to zero far away from the atom; we can think of it as being tied down, just as for a string tied down on both ends and free to move in between. In such circumstances, the tied-down string can only perform a discrete set of vibrations; likewise, the wave function of the electron has a discrete set of energy levels. Anytime the wave function of a system is “tied down” by going to zero for large/faraway/extreme configurations, it will exhibit a set of discrete energy levels.

Returning to field theory, consider a very simple field configuration, a sine wave stretching throughout all of space. We call such a configuration a *mode* of the field; it’s a convenient way of thinking, since any field configuration at all can be thought of as a combination of many modes of different wavelengths. That sine wave contains energy, and the energy increases rapidly as we imagine waves of greater and greater height. We want to construct the quantum wave function *of* that field. Because the energy of the field rises with the height of the wave, the wave function needs to decrease rapidly as the height of the wave increases, so as to not give too much probability to very high-energy waves. For all intents and purposes, the wave function is tied down (it goes to zero) at large energies.

As a result, just like a vibrating string or an electron in an atom, there is a discrete set of energy levels for the vibrations of a quantum field. In fact, every mode of the field can be in its lowest-energy state, or its next-highest, or next-highest, and so on. The overall minimum-energy wave function is